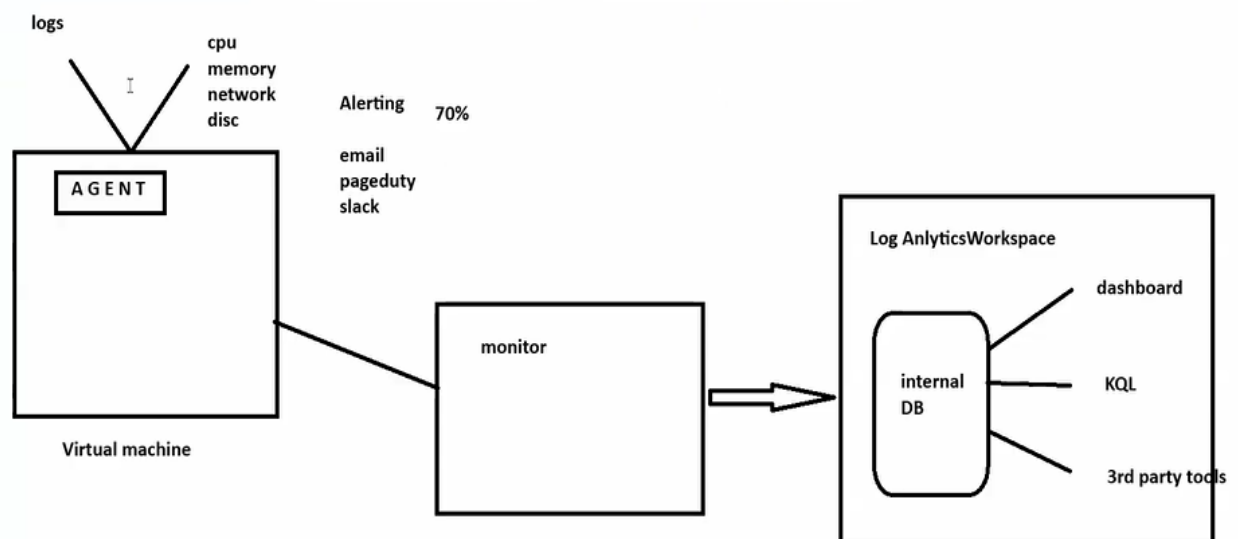


Azure Monitoring

✓ To maintain health status of our Machines we use Monitor

- ✓ When we enable monitor on this machine one agent will be installed on this machine.
- ✓ This agent responsible is to monitor this machine
- ✓ AGENT will collect the Metrics like (CPU,Memory,Network,Disc)
- ✓ At the same time, it will check LOGs like (Critical,Audit,Security)
- ✓ Once it is monitor, we can give Alerting,Pageduty,Slack as well
 - Example: if CPU percentage is 70%, we should get Alert on my Email.
- ✓ In this way we use monitor to our whole infrastructure and we mention above counters (CPU,MemoryNetwork,etc) based on our requirement and we can give Alerting to our Mails Directly.
- ✓ In real-time we cannot monitor individual machine bcz there are thousands of machines for that we use tools like (AzureMonitor,Promotus)



- When we enable monitoring, One Agent will be created on virtual machine and it will check all logs and send back to monitor, also it will store all the data into “Log Analytics Workspace” inside it has “Internal DB”.
- To trigger Internal DB we use Dashboards, KQL, 3rd party tools
- In this way we could able to monitor.

✓ Create VM

✓ Next search for log analytics workspace and create

Create Log Analytics workspace

With Azure Monitor Logs you can easily store, retain, and query data collected from your monitored resources in Azure and other environments for valuable insights. A Log Analytics workspace is the logical storage unit where your log data is collected and stored.

Project details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription *	Azure subscription 1
Resource group *	Urg01
	Create new

Instance details

Name *	Ulogworkspace1
Region *	South Central US

[Review + Create](#)[« Previous](#)[Next : Tags >](#)

✓ Next search for Monitor

Home



Monitor | Overview

Microsoft

Search

Overview

Activity log

Alerts

Issues (preview)

Metrics

Logs

Change Analysis

Service health

Workbooks

Dashboards with Grafana

Insights

Managed Services

Managed Prometheus



The Log Analytics agents, used by VM Insights, won't be supported as of August 31, 2024. Plan to migrate to VM Insights on Azure Monitor agent prior to this

Overview

Tutorials

Insights

Use curated monitoring views for specific Azure resources. [View all insights](#)



Application insights

Monitor your app's availability, performance, errors, and usage.



View ... More



Container Insights

Gain visibility into the performance and health of your controllers, nodes, and containers.



View ... More



VM Insights

Monitor the health, performance, and dependencies of your VMs and VM scale sets.



View ... More



Network Insights

View the health and metrics for all deployed network resources.



View ... More

✓ Click on VM insights View above

✓ Click on Configure Insights below

Home > Monitor



Monitor | Virtual Machines

Microsoft

Search



Refresh



Provide Feedback

Overview

Activity log

Alerts

Issues (preview)

Metrics

Logs

Change Analysis

Service health

Workbooks

Dashboards with Grafana

Insights

Applications

Virtual Machines

Storage accounts

Containers

Network

Get started

Overview

Performance

Map

Monitor the health and performance of virtual machines

VM insights monitors the performance and health of your virtual machines and virtual machine scale sets, including their running processes and dependencies on other resources. It can help deliver predictable performance and availability of vital applications by identifying performance bottlenecks and network issues. [Learn more](#)



Enable VM Insights

Implement complete monitoring of your Azure and hybrid virtual machine environment. [Learn more](#)

Configure Insights



Analyze data

Analyze the health and performance for a single machine or multiple machines and drill into logs for troubleshooting. [Learn more](#)

Analyze data



Create alerts

Alerts in Azure Monitor proactively identify interesting data and patterns in data and potentially take action on triggers. [Learn more](#)

✓ Enable Monitoring on VM below, then Agent will install on that server

Home > Monitor

Monitor | Virtual Machines

Microsoft

Search Refresh Provide Feedback

Overview Activity log Alerts Issues (preview) Metrics Logs Change Analysis Service health Workbooks Dashboards with Grafana Insights Applications **Virtual Machines** Storage accounts

Get started **Overview** Performance Map

Filter by name... Subscription: **Azure subscription 1** Resource group: **All resource groups** Type

Group by: **Subscription, Resource group**

Onboarding now supports OpenTelemetry metrics. If you enable only OpenTelemetry metrics, your VM will still collect data but will not appear as 'Monitored' in this view. [Learn more](#)

Monitored (0) **Not monitored (2)** Workspace configuration Other onboarding options

Name	Monitor Coverage
▼ Azure subscription 1	2 of 2
> bh-vpn-rg	1 of 1
▼ urg01	1 of 1
Uvm01	Not enabled Enable

Configure monitor | Uvm01

Capabilities Review + enable

Infrastructure monitoring

Collect health and performance data from the operating system running on your virtual machine for improved troubleshooting, alerts, and visualizations.

[Customize infrastructure monitoring](#)

Enable detailed metrics ⓘ

☐ [Preview] OpenTelemetry metrics [See metrics](#) ⓘ **At no additional cost**

☒ [Classic] Log-based metrics ⓘ
Log Analytics workspace: [ulogworkspace1](#)

✓ Below, VM in monitor state

Home

Monitor | Virtual Machines

Microsoft

Search Refresh Provide Feedback

Overview Activity log Alerts Issues (preview) Metrics Logs Change Analysis Service health Workbooks Dashboards with Grafana Insights Applications **Virtual Machines**

Get started **Overview** Performance Map

Filter by name... Subscription: **Azure subscription 1** Resource group: **All resource groups** Type: **All types**

Location: **All locations** Group by: **Subscription, Resource group**

Onboarding now supports OpenTelemetry metrics. If you enable only OpenTelemetry metrics, your VM will still collect data but will not appear as 'Monitored' in this view. [Learn more](#)

Monitored (1) Not monitored (1) Workspace configuration Other onboarding options

Name	Monitor Coverage	Data collection rule
▼ Azure subscription 1	1 of 2	
▼ urg01	1 of 1	
Uvm01	Enabled	MSVM-southcentralus-uum01

✓ Below we can see **Agent is installed** on VM below.

Home > Compute infrastructure | Virtual machines > Uvm01

Uvm01 | Extensions + applications

Virtual machine

Search

What extensions can help me keep my virtual machine secure? What is the difference between VM applications and extensions? What are the types of VM extensions available?

Tags

Diagnose and solve problems

Resource visualizer

Connect

Networking

Settings

Disks

Extensions + applications

Operating system

Extensions VM Applications

+ Add Refresh Update Enable automatic upgrade Disable automatic upgrade Feedback

Search to filter items...

Showing all 1 items

Name	Type	Version	Latest Version	Status	Automatic up
AzureMonitorWindowsA...	Microsoft.Azure.Monitor...	1.41.0.0	1.41.0.0	Provisioning succeeded	Enabled

✓ If we go down in Monitoring → Diagnostic settings here whatever the data on Vm it will store on storage account, below select storage account after creating it.

Home > Compute infrastructure | Virtual machines > Uvm01

Uvm01 | Diagnostic settings

Virtual machine

Search

Locks

Availability + scale

Security

Backup + disaster recovery

Operations

Monitoring

Insights

Alerts

Metrics

Diagnostic settings

Logs

Workbooks

Azure Diagnostics extension will be deprecated on March 31, 2026. After this date, Microsoft will no longer provide support for the Azure

Install Diagnostics Extension

Azure Monitoring collects host-level metrics - like CPU utilization, disk and network usage - for all virtual machines without any additional software. For more insight into this virtual machine, you can collect guest-level metrics, logs, and other diagnostic data using the Azure Diagnostics agent. You can also send diagnostic data to other services like Application Insights. [Learn more](#)

To get started now, choose a storage account below where diagnostic data will be sent and then click the button labeled 'Enable guest-level diagnostics'.

In portal, the storage account must be in the same region as the virtual machine. Your VM is located in 'southcentralus'.

Diagnostics storage account *

Select a storage account

Enable guest-level monitoring

✓ Next create **Data Storage**

Home

Storage center | Blob Storage

Default Directory (reddylkp2004gmail.onmicrosoft.com)

Generate cost-saving LCM rules

Search

Summary Resources

Overview

All storage resources

Object storage

File storage

+ Create Restore Manage view Refresh

Filter for any field...

Subscription equals all

Create a storage account ...

Subscription *

Azure subscription 1

Resource group *

Urg01

Create new

Instance details

Storage account name * ⓘ

ustorageacc

Region * ⓘ

(US) South Central US

Deploy to an Azure Extended Zone

Preferred storage type

Choose preferred storage type

This helps us provide relevant guidance. It doesn't restrict your storage to this resource type. [Learn more](#)

Performance * ⓘ

Standard: Recommended for most scenarios (general-purpose v2 account)

Premium: Recommended for scenarios that require low latency.

Redundancy * ⓘ

Locally-redundant storage (LRS)

- Basics
- Advanced
- Networking
- Data protection
- Encryption
- Tags
- Review + create

Security

Configure security settings that impact your storage account.

- Require secure transfer for REST API operations ⓘ
- Allow enabling anonymous access on individual containers ⓘ
- Enable storage account key access ⓘ

Your deployment is complete

	Deployment name : ustorageacc_1771504252654	Start time : 19/02/2026, 18:01:05
	Subscription : Azure subscription 1	Correlation ID : 3deac3f7-03fb-4e93-9006-4184b1bc9677
	Resource group : Urg01	

> Deployment details

✓ Next steps

Go to resource

✓ Below we can see storage account, select it.

Home > Compute infrastructure | Virtual machines > Uvm01

Uvm01 | Diagnostic settings ☆ ...
Virtual machine

Search

- Locks
- Availability + scale
- Security
- Backup + disaster recovery
- Operations
- Monitoring
 - Insights
 - Alerts
 - Metrics
 - Diagnostic settings**
 - Logs
 - Workbooks

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Diagnostics storage account *

Select a storage account

Urg01

ustorageacc

Uvm01 | Diagnostic settings ☆ ...
Virtual machine

Search

- Settings
 - Disks
 - Extensions + applications
 - Operating system
 - Configuration
 - Advisor recommendations
 - Properties
 - Locks
- Availability + scale
- Security
- Backup + disaster recovery
- Operations
- Monitoring
 - Insights
 - Alerts
 - Metrics
 - Diagnostic settings**
 - Logs

ⓘ Azure Diagnostics extension will be deprecated on March 31, 2026. After this date, Microsoft will no longer provide support for the Azure Diagnostics extension. [Learn more](#)

Agent Performance Counters Logs Event Tracing Crash Dumps Sinks

Windows Virtual Machine Diagnostics Extension

Azure Diagnostics extension is an agent in Azure Monitor that collects monitoring data from the guest operating system of Azure compute resources including virtual machines. Data is always collected into an Azure Storage Account, however you may configure one or more data sinks to send data to other destinations. [Learn more about the Windows Diagnostics Agent.](#)

Azure Diagnostics Agent Settings

Configure the settings for the diagnostics agent itself. [Learn more](#)

Storage account * ⓘ

Disk quota (MB): * ⓘ

Collect Infrastructure Logs ⓘ ☒

Log level: * ⓘ

Remove Azure Diagnostics agent

If diagnostic data isn't being collected or you're having trouble viewing it in the portal, reinstalling the agent might help. This removes the

✓ Click on performanceCounters and select basic counters whatever you need, below

Uvm01 | Diagnostic settings

Virtual machine

Search

Disks

Extensions + applications

Operating system

Configuration

Advisor recommendations

Properties

Locks

Availability + scale

Security

Backup + disaster recovery

Operations

Monitoring

Insights

Alerts

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Diagnostic settings

Logs

Workbooks

Automation

Help

Azure Diagnostics extension will be deprecated on March 31, 2026. After this date, Microsoft will no longer provide support for the Azure Diagnostics extension. [Learn more](#)

AgentPerformance CountersLogsEvent TracingCrash DumpsSinks

Windows Performance Counters

Windows Performance Counters provide a high-level abstraction layer to system data such as CPU, memory, and disk usage. System administrators use performance or behavior problems. Software developers often use performance counters to monitor their applications. [Learn more](#)

+ Select basic counters

+ Add custom counters

✕ Clear all counters

Counter specifier

\\Processor Information(_Total)\\% Processor Time

\\Processor Information(_Total)\\% Privileged Time

\\Processor Information(_Total)\\% User Time

\\Processor Information(_Total)\\Processor Frequency

\\System\\Processes

\\Process(_Total)\\Thread Count

\\Process(_Total)\\Handle Count

Apply

Discard changes

Select basic counter groups

Select basic performance counter groups. Selecting a group will enable a number of performance counters for your application.

① Selecting basic counter groups will reset existing changes to your basic performance counters. Your custom counters will not be affected.

Select basic counters * ①

Sample rate * ①

CPU (10), Disk (18), Memory (10), Network (8), ASP.NET (140), SQL Server (12)

☒ CPU (10)

☒ Disk (18)

☒ Memory (10)

☒ Network (8)

☒ ASP.NET (140)

☒ SQL Server (12)

Select

Cancel

✓ **Logs below**

Uvm01 | Diagnostic settings

Virtual machine

Search

Disks

Extensions + applications

Operating system

Configuration

Advisor recommendations

Properties

Locks

Availability + scale

Security

Backup + disaster recovery

Operations

Monitoring

Insights

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Diagnostic settings

Logs

Workbooks

Automation

Help

Azure Diagnostics extension will be deprecated on March 31, 2026. After this date, Microsoft will no longer provide support for the Azure Diagnostics extension. [Learn more](#)

AgentPerformance Counters**Logs**Event TracingCrash DumpsSinks

Event logs

Event logging provides a standard, centralized way for applications (and the operating system) to record important software and hardware events. The event logging service records events from various sources and stores them in a single collection called an event log. [Learn more](#)

Collect Event Logs ①

Basic event logs to collect ①

Add additional custom event logs ①

✕ Clear custom event logs

Configured custom event logs

Failed Requests

Failed Request Tracing is a powerful diagnostics feature helps developers and administrators detect application errors and why they are happening. [Learn more](#)

Collect failed request logs ①

IIS Logs

Apply

Discard changes

Application logs

☒ Application critical

☒ Application error

☒ Application warning

☐ Application information

☐ Application verbose

Security logs

☒ Audit success

☒ Audit failure

System logs

☒ System critical

☒ System error

☒ System warning

☐ System information

☐ System verbose

- ✓ Incase Vm crashes that data will store on Crash dumps, based on issue we can work on it.

Uvm01 | Diagnostic settings

Virtual machine

Search

Settings

Disks

Extensions + applications

Operating system

Configuration

Advisor

Azure Diagnostics extension will be deprecated on March 31, 2026. After this date, Microsoft will no longer provide support for the Diagnostics extension. [Learn more](#)

Agent Performance Counters Logs Event Tracing **Crash Dumps** Sinks

Crash Dumps

Collect memory dumps when a process crashes. If no processes have been specified, this will do nothing. [Learn more](#)

Collect Crash Dumps ☐

- ✓ Sinks Enable it and apply below

Uvm01 | Diagnostic settings

Virtual machine

Search

Settings

Disks

Extensions + applications

Operating system

Configuration

Advisor recommendations

Properties

Locks

Azure Diagnostics extension will be deprecated on March 31, 2026. After this date, Microsoft will no longer provide support for the Diagnostics extension. [Learn more](#)

Agent Performance Counters Logs Event Tracing Crash Dumps **Sinks**

Send your diagnostic data to other services for more insights. Additional charges may apply.

Azure Monitor (Preview)

Azure Monitor helps you maximize performance and availability of your applications and proactively identify problems by collecting, analyzing, and acting on telemetry. Enabling this option will send diagnostic data collected by the Windows Diagnostics (WAD) agent to Azure Monitor, but it will not install the Azure Monitor agent itself on your VM. [Learn more](#)

Enable Azure Monitor ☒

- Login into VM using RDC
- Next open Metrics on VM

Home > Compute infrastructure | Virtual machines > Uvm01

Uvm01 | Metrics

Virtual machine

What metrics do people commonly use to track this kind of resource? What metrics are available for this resource? Can I get alerted based on live metrics?

Search

+ New chart Refresh Share

Local Time: Last 24 hours (Auto)

Chart Title

+ Add metric Add filter Apply splitting

Line chart Drill into Logs New alert rule Save to dashboard

Scope	Metric Namespace	Metric	Aggregation
Uvm01	Virtual Machine Host	cpu	Select aggregation

CPU Credits Consumed

CPU Credits Remaining

Percentage CPU

Enable guest memory metrics

Select a metric

Filter + Split

Plot multiple metrics

Build custom dashboards

Percentage CPU

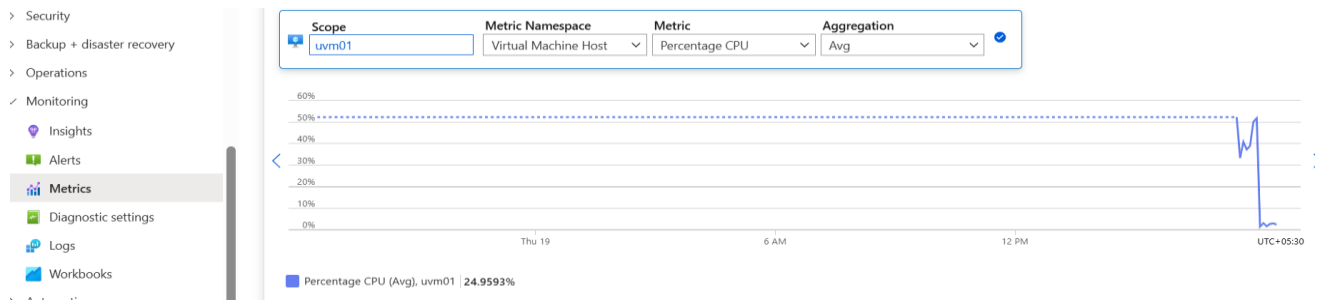
The percentage of allocated compute units that are currently in use by the Virtual Machine(s)

Metric ID: Percentage CPU

Namespace: microsoft.compute/virtualmachines

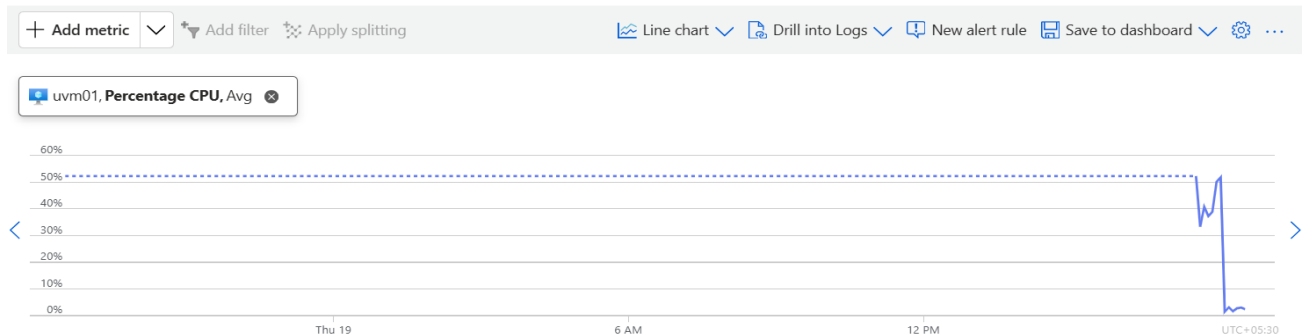
Unit: %

✓ We can see metric here and save as Dashboard



✓ Click on **New alert rule** below

Avg Percentage CPU for uvm01



✓ Give below rules, using **percentage CPU**(Counter)

Create an alert rule ...

Scope Condition Actions Details Tags Review + create

Configure when the alert rule should trigger by selecting a signal and defining its logic.

Signal name * Percentage CPU

[See all signals](#)

Alert logic

Threshold type ☒ Static ☐ Dynamic

Aggregation type Maximum

Value is Greater than or equal to

Threshold * 40 %

When to evaluate

Check every 1 minute

Lookback period 1 minute

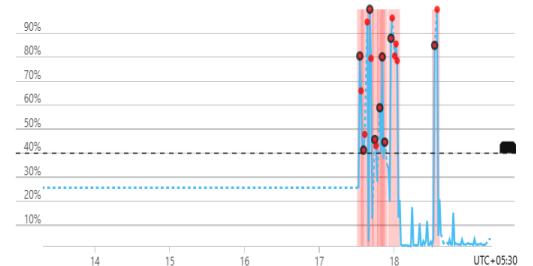
Preview

\$0.10 USD/mo

Whenever the maximum Percentage CPU is greater than or equal to 40%

Preview time range : **Over the last 6 hours**

Time series : **Aggregate**



[Review + create](#)

[Previous](#)

[Next: Actions >](#)

✓ Click on **actions** and **create group**

Create an alert rule ...

Scope Condition **Actions** Details Tags Review + create

An action group is a set of actions that can be applied to an alert rule. [Learn more](#)

+ Select action groups + Create action group

Action group name

Contains actions

No action group selected yet

Home > Compute infrastructure | Virtual machines > Uvm01 | Metrics > Create an alert rule

Create action group ...

Basics Notifications Actions Tags Review + create

An action group invokes a defined set of notifications and actions when an alert is triggered. [Learn more](#)

Project details

Select a subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all

Subscription ⓘ

Azure subscription 1

Resource group * ⓘ

Urg01

[Create new](#)

Region *

Global

Instance details

Action group name * ⓘ

emailaction

Display name * ⓘ

emailaction

The display name is limited to 12 characters

Review + create

Previous

Next: Notifications >

✓ **Select**

Create action group ...

Basics **Notifications** Actions Tags Review + create

Choose how to get notified when the action group is triggered. This step is optional.

Notification type ⓘ

Name ⓘ

Selected ⓘ

▼

Email Azure Resource Manager Role

Email/SMS message/Push/Voice

✓ Select your convenient Alert below

Home > Compute infrastructure | Virtual machines > Uvm01 | Metrics > Create an alert rule

Create action group ...

Basics **Notifications** Actions Tags Review + create

Choose how to get notified when the action group is triggered. This step is optional.

Notification type ⓘ	Name ⓘ	Selected ⓘ
Email/SMS message... ▾		
▾		

Review + create

Previous

Next: Actions >

✓ Review +Create

Create action group ...

Basics Notifications Actions Tags **Review + create**

This is a summary of your action group. Please review to ensure the information is correct and consistent. [Statement](#).

Basics

Subscription	Azure subscription 1
Resource group	Urg01
Region	global
Action group name	emailaction
Display name	emailaction

Notifications

Notification type	Name	Selected
Email/SMS message/Push/Voice	EmailAlertUday	Email

✓ If we want any other conditions select on conditions below and create action group.

Email/SMS message/Push/Voice

Add or edit Email/SMS message/Push/Voice action

☒ Email
Email * ⓘ udayreddychalla10@gmail.com

☒ SMS (Carrier charges may apply)
Country code 1
Phone number

☐ Azure mobile app notification
Azure account email ⓘ

☐ Voice
Country code 1
Phone number

Enable the common alert schema. [Learn more](#)

Yes No

OK

Create an alert rule ...

Scope Condition Actions Details Tags Review + create

An action group is a set of actions that can be applied to an alert rule. [Learn more](#)

[+ Select action groups](#) [+ Create action group](#)

Action group name	Contains actions
emailaction	1 Email

✓ **Next go to details below**

[Home](#) > [Compute infrastructure | Virtual machines](#) > [Uvm01 | Metrics](#)

Create an alert rule ...

Scope Condition Actions Details Tags Review + create

Project details

Select the subscription and resource group in which to save the alert rule.

Subscription * ⓘ

Azure subscription 1

Resource group * ⓘ

Urg01

[Create new](#)

Alert rule details

Severity * ⓘ

0 - Critical

Alert rule name * ⓘ

CPUgreaterthan70

Alert rule description ⓘ

CPUgreaterthan70

∨ **Advanced options**

✓ **Below Review + Create**

Home > Compute infrastructure | Virtual machines > Uvm01 | Metrics

Create an alert rule ...

Scope Condition Actions Details Tags Review + create

Product details

Metric alert rule	Total pricing
1 Condition	0.10 USD/month
Terms of use Privacy statement	Pricing

Scope

Scope level	🔑 Subscription
Resource	🔑 Azure subscription 1 > 🌐 urg01 > 🖥️ uvm01

Condition

Signal name	Percentage CPU
Operator	Greater than or equal to
Aggregation type	Maximum
Threshold value	40
Lookback period	1 minute
Check every	1 minute

Actions

Action group name	Contain actions
emailaction	1 Email ⓘ

Details

Project details

Subscription	Azure subscription 1
Resource group	Urg01
Region	global

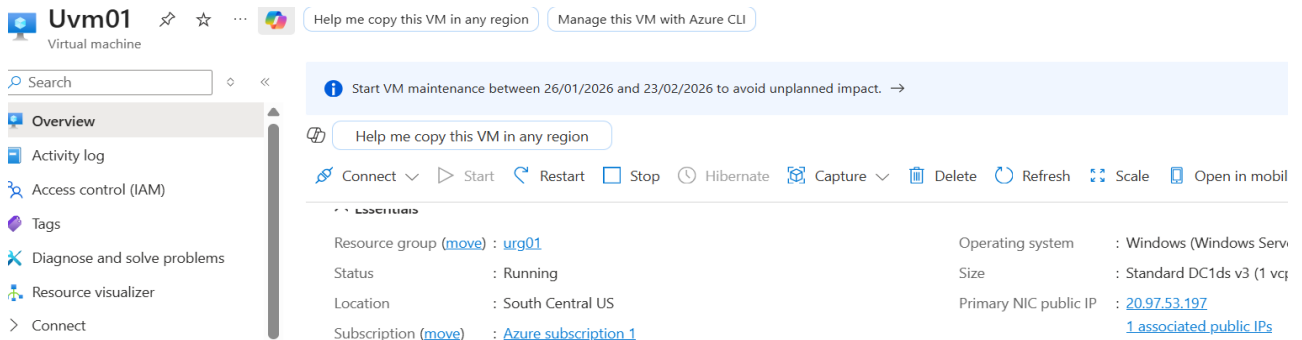
Alert rule details

Alert rule name	CPUgreaterthan70
Alert rule description	CPUgreaterthan70
Severity	🔴 0 - Critical
Enable upon creation	☒
Automatically resolve alerts	☒

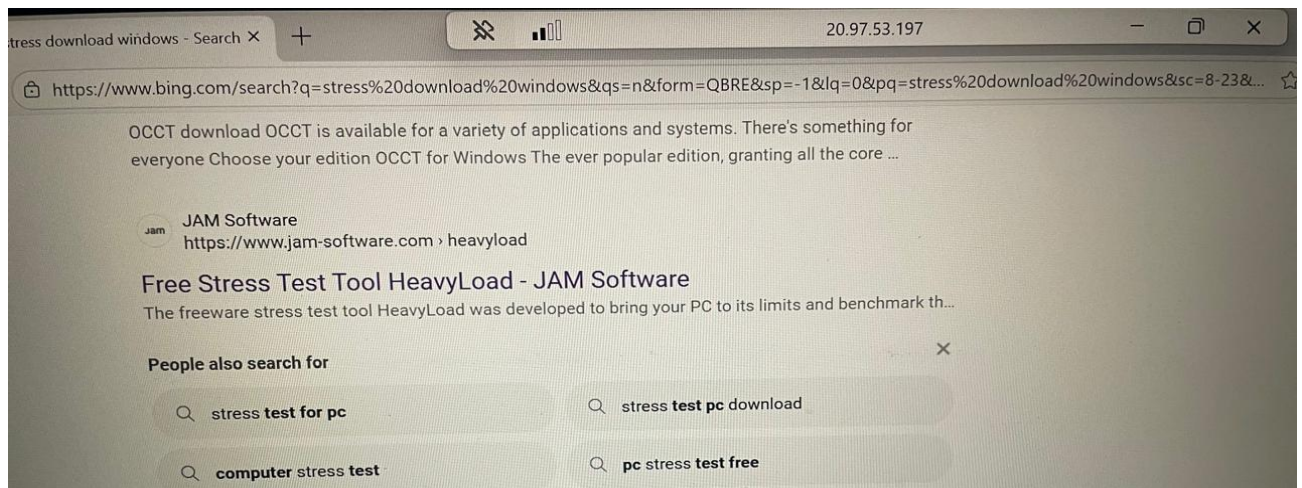
Create

Previous

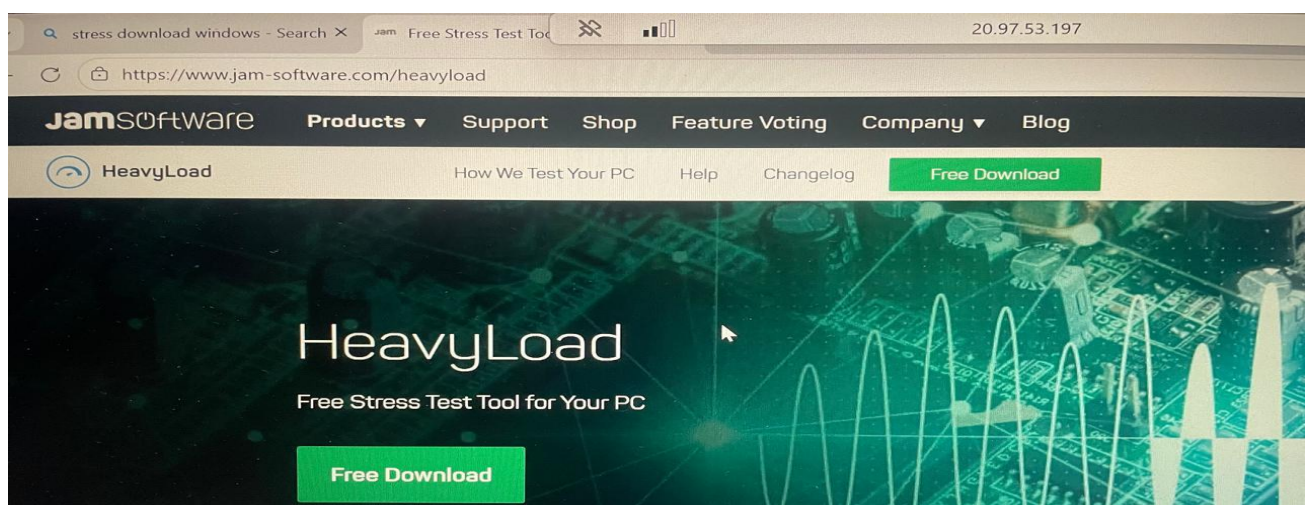
- ✓ Now download **STRESS** software on **VM**, which we logged in earlier.
- ✓ Stress app to increase traffic on VM server so that CPU will be 100% then Azure monitor will send alert to our respective mail address



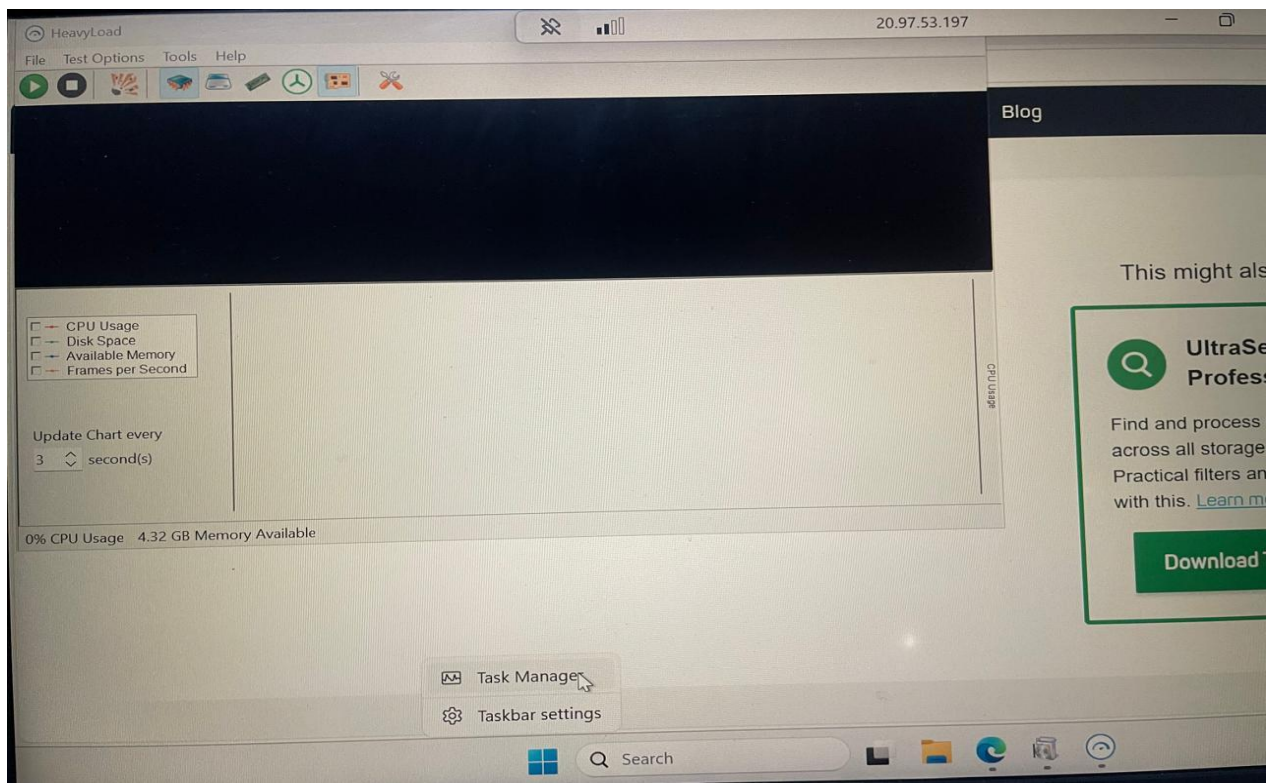
- ✓ Search for stress below



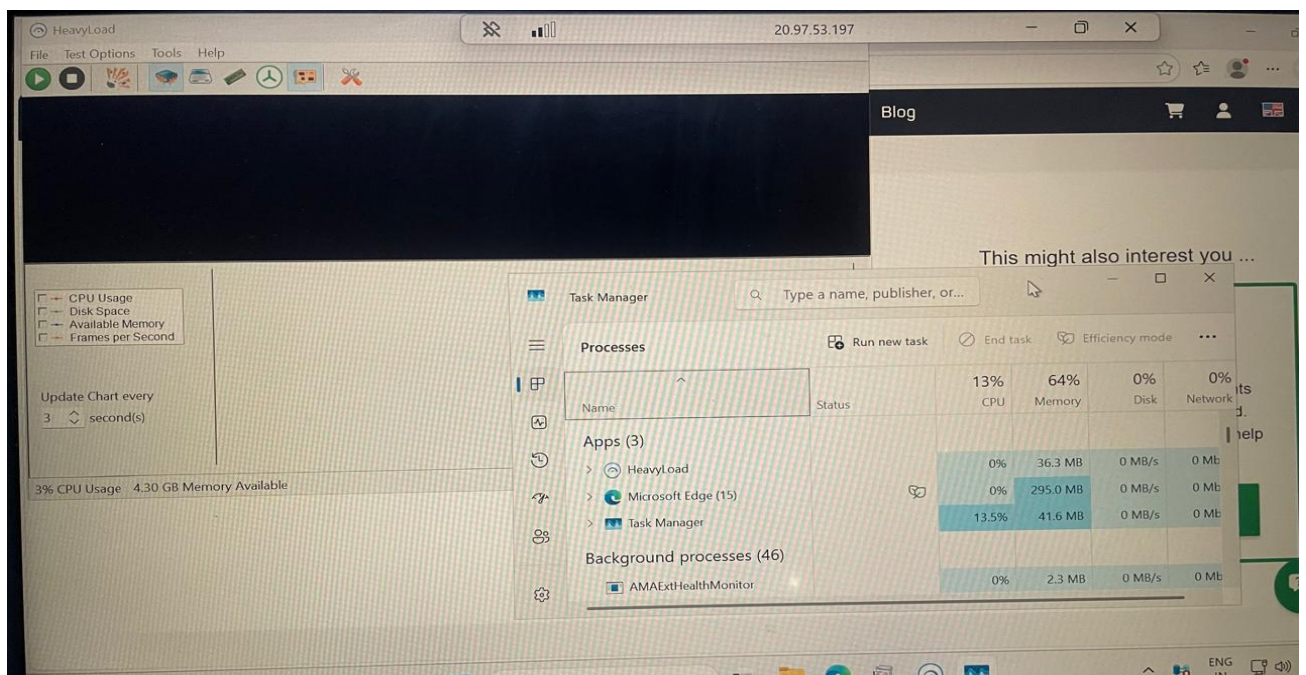
- ✓ Download below



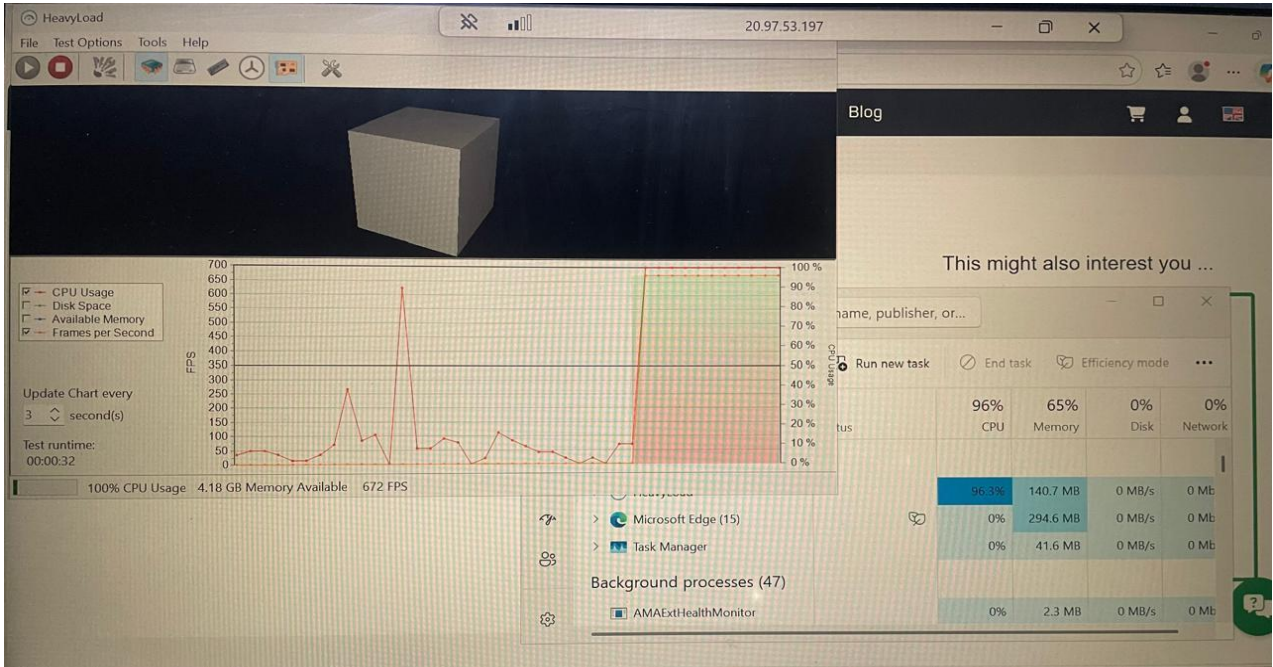
- ✓ Once we install it will look like below.
- ✓ To **monitor traffic** open **Task Manager** by Right-Click on **Tool bar** you can see below.



- ✓ Below traffic is below 20%
- ✓ Let's increase traffic by clicking on green button on stress



- ✓ Once traffic increase, CPU reaches 100%,
 - ✓ Azure Monitor agent continuously monitors the VM.
- When the traffic or metric reaches the configured threshold percentage, an alert is automatically triggered.



✓ Below alert was triggered.

 **Microsoft Azure** <azure-noreply@microsoft.com> [Unsubscribe](#) 7:59 PM (23 min)
to me ▾

 Microsoft Azure

Your Azure Monitor alert was triggered

Azure monitor alert rule CPUgreaterthan70 was triggered for Uvm01 at February 19, 2026 14:28 UTC.

Alert rule description	CPUgreaterthan70
Rule ID	/subscriptions/5d75b66d-66bf-44e0-8d7e-7e61e4b043d7/resourceGroups/Urg01/providers/microsoft.insights/metricAlerts/CPUgreaterthan70



Microsoft Azure <azure-noreply@microsoft.com>
to me ▾

8:15 PM (7 min)



 Your Azure Monitor alert was triggered

Azure monitor alert rule CPUgreaterthan70 was triggered for Uvm01 at February 19, 2026 14:44 UTC.

Alert rule description	CPUgreaterthan70
Rule ID	/subscriptions/5d75b66d-66bf-44e0-8d7e-7e61e4b043d7/resourceGroups/Urg01/providers/microsoft.insights/metricAlerts/CPUgreaterthan70 View Rule